

Image Sampling vs Quantization

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Image sampling and image quantization are two fundamental steps in converting real-world images into digital form. Sampling determines how many pixels are used to represent an image, while quantization decides how many intensity levels or colours each pixel can store. Together, they control the quality, clarity and file size of digital images.

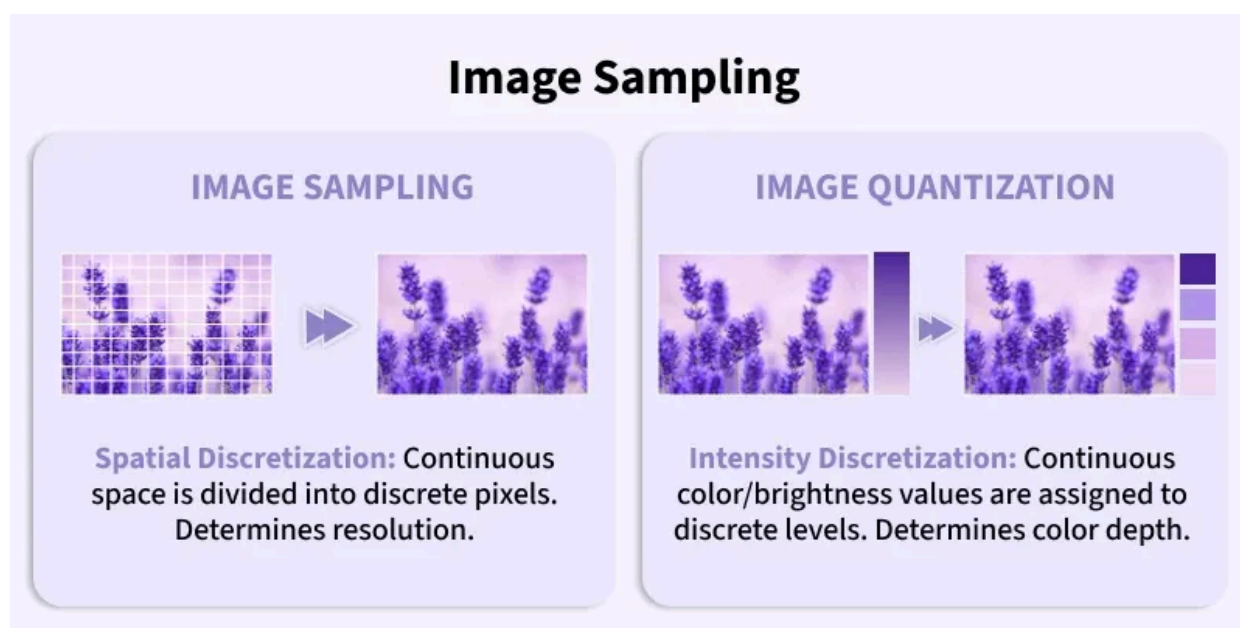


Image Sampling vs Quantization

1. Image Sampling

Image sampling is the process of converting a continuous image into a discrete form by selecting a finite number of points (pixels) from the image plane. It deals with spatial resolution i.e how many pixels are used to represent the image in horizontal and vertical directions. In simple terms, sampling determines where pixel values are taken from the image.

- Converts continuous spatial coordinates into a discrete grid.
- Directly controls image resolution (e.g., 256×256, 512×512).
- Higher sampling rate means more pixels which means more detail.
- Lower sampling rate means fewer pixels leading to loss of spatial detail.
- Poor sampling can cause aliasing effects.

Working

- The continuous image is overlaid with a rectangular grid.
- Each grid intersection corresponds to a pixel location.
- The image intensity is measured at each grid point.

- The spacing between grid points defines the sampling rate.
- The sampled values form a 2D array of pixels.

Use Cases

- **Digital cameras:** Determining sensor resolution.
- **Medical imaging:** CT and MRI scan resolution control.
- **Satellite imagery:** Choosing ground spatial resolution.
- **Image resizing:** Downsampling or upsampling images.

Example

- A real-world scene is sampled into a 1024×1024 pixel image.
- Reducing it to 256×256 means fewer samples, resulting in blurred edges and loss of fine details.

2. Image Quantization

Image quantization is the process of mapping a large set of continuous or discrete intensity values to a smaller, finite set of intensity levels. It deals with intensity resolution i.e how many grey levels or colors are used to represent pixel values. In simple terms, quantization determines how accurately pixel values are represented.

- Converts continuous intensity values into discrete levels.
- Controls the number of grey levels or colors.
- More levels means smoother intensity transitions.
- Fewer levels means visible banding and distortion.
- Quantization introduces quantization error.

Working

- Pixel intensity values are first sampled.
- A finite set of intensity levels is defined (e.g., 8-bit → 256 levels).
- Each pixel value is rounded to the nearest allowed level.
- The difference between original and assigned value is the quantization error.
- The result is a digitally representable image.

Use Cases

- **Image storage:** Reducing memory requirements.
- **Image compression:** JPEG and PNG encoding.
- **Display systems:** Limited color depth screens.
- **Computer vision:** Simplifying intensity ranges for processing.

Example

- An image with intensities from 0–255 (8-bit) is quantized to 16 levels.
- Smooth gradients appear as visible steps (posterization effect).

Relationship Between Image Sampling and Image Quantization

- Image sampling and image quantization are complementary steps in the digitization of a continuous image and both must be applied to obtain a complete digital representation.
- Sampling discretizes the spatial domain of an image by deciding the number and arrangement of pixels along the horizontal and vertical axes, thereby determining the image resolution and structural detail.
- Quantization discretizes the intensity domain by assigning each sampled pixel a finite set of intensity or color levels, thereby controlling brightness accuracy and tonal smoothness.
- Inadequate sampling results in spatial distortion, such as aliasing and loss of fine details, even if intensity values are represented accurately.
- Inadequate quantization results in intensity distortion, such as banding and false contours, even if the spatial resolution is high.
- High-quality digital images require both sufficient sampling density and sufficient quantization levels, as improving one cannot compensate for deficiencies in the other.

Comparison

Let's compare them:

Aspect	Image Sampling	Image Quantization
Basic Meaning	Process of selecting discrete spatial points (pixels) from a continuous image	Process of mapping pixel intensity values to a finite set of levels
Domain of Operation	Spatial domain (x, y coordinates)	Intensity / amplitude domain
Controls	Image resolution and level of spatial detail	Gray level or color resolution
Key Parameter	Sampling rate or number of pixels	Bit depth or number of intensity levels
Main Information Loss	Loss of fine spatial details and aliasing	Loss of intensity precision and false contours
Visual Impact	Affects image sharpness and structure	Affects image smoothness and tonal transitions